SIMATS ENGINEERING

Department of B.Tech Biomedical Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code: CSA17 **Course Title:** Artificial Intelligence

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CO Assessed	CO1 – Apply (BL3)

QUESTION 1 – TWO-ROOM VACUUM WORLD

The vacuum world has two squares, A and B. The agent starts at A and both squares are initially dirty. The performance measure is +1 for every clean square at the end of each time step, with a penalty of p for every Left or Right movement.

(a) PEAS Description

Component	Description
Performance Measure	+1 for every clean square at the end of each step; penalty p for every movement.
Environment	Two squares A and B; initially both are dirty.
Actuators	Suck, Left, Right, NoOp.
Sensors	Perceives current location and whether the current square is Dirty or Clean.

(b) Agent X – Simple Reflex Agent

Step	Location	Percept	Action	Clean squares	Penalty	Score
1	A	Dirty	Suck	A	0	+1
2	A	Clean	Right	A	1	0
3	B	Dirty	Suck	A, B	0	+2
4	B	Clean	Left	A, B	1	+1
5	A	Clean	Right	A, B	1	+1
6	B	Clean	Left	A, B	1	+1

Total score = 1 + 0 + 2 + 1 + 1 + 1 = 6 points.

(c) Agent Y – Reflex with Internal State

Step	Location	Percept	Action	Clean squares	Penalty	Score
1	A	Dirty	Suck	A	0	+1
2	A	Clean	Right	A	1	0
3	B	Dirty	Suck	A, B	0	+2

Step	Location	Percept	Action	Clean squares	Penalty	Score
4	B	Both clean remembered	NoOp	A, B	0	+2
5	B	Both clean remembered	NoOp	A, B	0	+2
6	B	Both clean remembered	NoOp	A, B	0	+2

Total score = 1 + 0 + 2 + 2 + 2 + 2 = 9 points.

(d) Score as a function of p

$S_X = 10 - 4p$
$S_Y = 10 - p$
Y is strictly better when $10 - p > 10 - 4p$, giving **$p > 0$**. At $p = 0$, both are equal.

(e) Rationality

Agent Y is more rational under the given performance measure because it uses percept history to remember that both squares are clean and then avoids unnecessary movement. Agent X reacts only to the current percept and therefore continues moving. Thus, for $p > 0$, Y achieves the higher performance.

QUESTION 2 – TASK ENVIRONMENT ANALYSIS AND AGENT SELECTION

(a) Classification of the three environments

Property	E1 – Autonomous Taxi	E2 – Part-Picking Robot	E3 – Chess
Observability	Partially observable – incomplete traffic information	Partially observable – camera limitations	Fully observable – board state visible
Agents	Multi-agent – other vehicles/drivers	Single agent	Multi-agent – two players
Determinism	Stochastic – traffic changes	Stochastic – sensing/conveyor uncertainty	Deterministic – fixed rules
Episodic/Sequential	Sequential	Sequential	Sequential
Static/Dynamic	Dynamic	Dynamic	Semi-dynamic – clock changes
Discrete/Continuous	Continuous	Continuous	Discrete

(b) Difficulty scores

E1: 1+1+1+1+1+1 = **6**
E2: 1+0+1+1+1+1 = **5**
E3: 1+1+0+1+0.5+0 = **3.5**

Ranking: E1 > E2 > E3.

(c) Agent architecture

Hardest – E1: Utility-based agent. The taxi needs to handle partial observability, multiple agents, uncertainty, dynamic conditions, continuous changes and sequential consequences.

Easiest – E3: Goal-based agent. Chess is fully observable, discrete and governed by deterministic rules, while the goal is to achieve checkmate.

(d) Interpretation

The environment properties determine the complexity of the agent program. A dynamic, stochastic, partially observable and continuous environment such as autonomous driving requires more sophisticated reasoning than a simpler discrete environment. Therefore, agent design depends on the environment, not merely its domain.

QUESTION 3 – TURING TEST DATA ACROSS THE HISTORY OF AI

(a) Correct-identification and fooling rates

System	Interrogations	Correctly identified	Correct ID rate	Fooling rate
ELIZA	50	47	94%	6%
Chatbot B	50	36	72%	28%
System C	50	29	58%	42%

Formula: Correct-identification rate = (correct identifications / total interrogations) × 100. Fooling rate = 100 – correct-identification rate.

(b) Turing 70% criterion

ELIZA: 94% > 70%, so it does **not** satisfy the criterion; it is 44 percentage points above the 50% guessing level.

Chatbot B: 72% > 70%, so it does **not** satisfy the criterion; it is 22 percentage points above the 50% level.

System C: 58% < 70%, so it **satisfies** the criterion; it is 8 percentage points above the 50% level.

(c) Improvement in fooling rate

ELIZA → System C: 42% – 6% = **36 percentage points**. Ratio = 42/6 = **7×**.

Chatbot B → System C: 42% – 28% = **14 percentage points**.

(d) Interpretation

The Turing Test evaluates whether a machine can act humanly in conversation. System C's 42% fooling rate does not prove that it is thinking or acting rationally. Human-like conversational behaviour can result from learned patterns without demonstrating human-like understanding or rationality. Thus, acting humanly is not necessarily the same as thinking humanly or acting rationally.

QUESTION 4 – RATIONAL ROUTE CHOICE UNDER UNCERTAINTY

Utility function: **U = 100** – **T** – **20L**, where L = 1 when the taxi arrives after 55 minutes, otherwise L = 0.

(a) Utility in each outcome

Route	Outcome	Time	Late?	Utility
R1	Normal	30 min	No	70
R1	Disrupted	70 min	Yes	10
R2	Normal	45 min	No	55
R2	Disrupted	60 min	Yes	20
R3	Normal	50 min	No	50

(b) Expected utility

R1: EU = (0.70)(70) + (0.30)(10) = **52**.

R2: EU = (0.90)(55) + (0.10)(20) = **51.5**.

R3: EU = 1(50) = **50**.

Therefore, the rational choice is **R1 – Highway**, because it has the highest expected utility.

(c) New highway accident alert

New disruption probability for R1 = 0.80; normal probability = 0.20.

EU(R1) = (0.20)(70) + (0.80)(10) = 14 + 8 = **22**.

Since R2 = 51.5 and R3 = 50, the new rational choice is **R2 – City Road**.

(d) Actual utility and rationality

If R1 was selected in part (b) and the disruption occurred, the actual utility is U = 100 – 70 – 20 = **10**. The agent was **not irrational**, because it selected R1 using the information and probabilities available before the disruption.

Rational agent: chooses the action with the best expected performance using available percepts. **Omniscient agent:** would know the actual future outcome. A rational agent is therefore not necessarily omniscient.

(e) Interpretation

Rationality depends on the percept sequence and performance measure. Initially R1 has the highest expected utility (52). After the radio alert, its disruption probability rises to 0.80 and its expected utility falls to 22, so R2 becomes preferable. The roads did not change; the agent's information changed.

New percept → new probabilities → new expected utility → new decision.

FINAL ANSWERS SUMMARY

Question	Final answer
Q1	X = 6, Y = 9 for p=1; Sx = 10–4p, Sy = 10–p; Y better for p>0.
Q2	E1 = 6, E2 = 5, E3 = 3.5; ranking E1 > E2 > E3.
Q3	ELIZA: 94%/6%; B: 72%/28%; C: 58%/42%; only C satisfies 70% criterion.
Q4	R1 = 52, R2 = 51.5, R3 = 50; initially R1; after alert R2; disrupted R1 utility = 10.